Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/16/2025
PI Name:	Hein, Travis Ph.D.		
Project Title:	Markers of Vascular Risks Associated with Spaceflight Associated Neuro-ocular Syndrome		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77807	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	04/01/2021	**End Date:**	03/31/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2027 per NSSC information (Ed., 1/4/22).		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Tharakan, Binu Ph.D. (Morehouse School of Medicine) Zawieja, David Ph.D. (Texas A&M University)		
Grant/Contract No.:	80NSSC21K0752		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight Associated Neuro-ocular Syndrome (SANS) is characterized by the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, or choroidal folds. It affects most astronauts completing long-duration spaceflights. The leading hypothesis for development of SANS is prolonged exposure to the headward fluid shift that occurs in weightlessness. While the exact pathophysiology of SANS remains unclear, since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we believe changes in ocular vascular hydrodynamics are likely to precipitate or contribute to development of SANS, particularly those findings that could be associated with localized edema. Appropriate perfusion of the optic nerve head and inner retina for delivery of oxygen and nutrients is dependent on ocular artery pressure gradients driving blood flow. Dysfunctional regulation of retinal blood flow or pressure may thus contribute to the SANS. Dysfunctions in other ocular vascular elements, venous and lymphatic vessels (ocular venous congestion and/or elevated venous compliance, disruption of blood-retina/brain barriers, and/or reduction in ocular lymph flow), may also contribute to optic disc edema, choroidal folds, and retinal layer thickening seen in SANS. There has been no systematic analysis of associated biomarkers of ocular vascular dysfunction. We have assembled a team of experts in all 3 main vascular types (arteries, veins, and lymphatics) to address this information gap. Thus, the objective of this application is to determine what biomarkers of spaceflight-associated vascular dysfunction might be useful to monitor and address SANS. The central hypothesis of this proposal is that spaceflight-induced dysfunction in the ocular vasculature leads to symptoms of SANS and will be associated with markers of vascular dysfunction that can be found in the cerebrospinal fluid (CSF) of affected astronauts. We will accomplish this by defining and measuring biomarkers in the CSF that are associated with ocular vascular dysfunction (artery and vein vasomotor regulation, venular permeability, and lymphatic transport function). We will focus our efforts to identify vascular biomarkers of: 1) ocular vascular vasomotor function (endothelin [ET-1], calcitonin gene-related peptide [CGRP], neuropeptide Y [NPY], and substance P [SP]); and 2) vascular inflammation (inflammatory cytokines and products). ET-1 and NPY are potent vasoconstrictors; CGRP causes vasodilation; and ET-1 and SP alter lymphatic pump dynamics. Elevated levels of these 4 vasoactive proteins have been detected in the CSF under stress conditions associated with inflammation. Edema and impaired lymphatic function are associated with increased localized inflammation and inflammatory products. Many inflammatory mediators are well-known modulators of vascular permeability and lymphatic transport function (leukotrienes, SP, interleukins). Thus, these may be important markers of vascular inflammation that could be associated with SANS. We will measure levels of the vasomotor and inflammatory biomarkers in astronaut CSF samples via ELISA and cytokine multiplex assays. These experiments will be conducted on CSF samples collected from participating International Space Station (ISS) crewmembers twice after their missions. Information from these studies will provide the first comprehensive analysis of biomarkers of ocular vascular dysfunction resulting from spaceflight that are associated with development of SANS in astronauts. It will help us understand the relationships between physiological and visual outcomes versus CSF biomarkers and thus lead to a better understanding of the mechanisms in SANS. This new knowledge regarding CSF biomarkers in spaceflight might also result in development of personalized countermeasures to protect ocular health and enhance visual performance during space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outcomes from this investigation may provide insight into vascular/inflammatory biomarkers to consider in evaluating and treating ocular diseases related to increased swelling and pressure in the eyes and brain of people on Earth, such as glaucoma and ocular/intracranial hypertension.
Task Progress:	During this fourth year of the grant, our study has recently been selected for the first time to be involved in sample collection after flight during a future mission. However, we have not obtained samples for the experiments yet. For the coming year, we will plan to conduct the experimental plan if samples are collected after flight.
Bibliography Type:	Description: (Last Updated: 07/07/2026)
Abstracts for Journals and Proceedings	Hein TW, Cromer WE, Rosa RH Jr, Tharakan B, Zawieja DC. "Markers of vascular risks associated with spaceflight associated neuro-ocular syndrome." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025